Intelligent Retail Decision Assistant - Capstone Report

Capstone Project · Arpit Jain · Run mode: LIVE (gpt-4o-mini + text-embedding-3-small)

1. Problem and approach

A mid-sized online fashion retailer needs one assistant that (a) explains and predicts **conversion** for merchandising, (b) segments customers for CRM, (c) recommends products grounded in the catalogue, (d) answers return / exchange / shipping / refund / warranty questions from policy documents only, and (e) routes every request to the right tool with a visible trace. Architecture: *tabular data → classical ML (scikit-learn) → retrieval-ready product & policy documents → embeddings + FAISS → grounded LLM generation with validators → rule-first hybrid controller*. Code: `src/`; walkthrough: notebooks 01-06.

2. Data and assumptions

- Course dataset: 1,500 interaction rows × 46 columns + 5 policies (retail_policies.json). 9 engineered columns added; full data dictionary in data/processed/data_dictionary.csv (original vs engineered, with how each was created).
- **Audit findings:** product_id and customer_id are not stable keys (each id spans 5 / 6 rows describing different products / people) → every row is treated as its own catalogue item (new item_id) and analysis is interaction-level. size is 18% missing at random → 'Unknown' + indicator. Post-purchase and checkout columns are excluded from the purchase model (leakage).
- Text columns (titles, descriptions, reviews, queries, tags) are template-generated in the source data; no policy text was invented. Image files are not supplied, so the optional multimodal extension was not attempted.

3. Supervised ML - predicting purchase_flag

Model	CV ROC-AUC (±std)	Test ROC-AUC	Test PR-AUC	Accuracy	Precision	Recall	F1
Dummy (majority)	0.500 ± 0.000	0.500	0.513	0.513	0.513	1.000	0.678
Logistic Regression	0.640 ± 0.027	0.678	0.670	0.660	0.662	0.688	0.675
Random Forest	0.641 ± 0.024	0.664	0.695	0.623	0.636	0.623	0.630
Gradient Boosting (HGB)	0.638 ± 0.029	0.676	0.696	0.630	0.629	0.682	0.654

Balanced target (51/49) → no resampling. Pipelines impute, scale and one-hot encode inside each CV fold; 80/20 stratified hold-out + 5-fold grid search. All models reach CV ROC-AUC ≈ 0.64 (gap < 1 std), so the interpretable **Logistic Regression (L1, C=0.1)** is chosen (test ROC-AUC 0.678). Permutation importance: **membership_ord, income_ord, avg_product_rating, engagement**. Membership tier dominates (Gold/Platinum ≈ 61-63% conversion vs ≈ 40% Bronze/Silver); price and discount barely matter on average. The signal is modest - useful for prioritising nudges, not for certain individual decisions.

4. Unsupervised ML - RFM customer segments

K-Means on standardised Recency (days since last purchase), Frequency (orders) and Monetary value (avg order value); k = 4 chosen by silhouette (≈ 0.26; a 9-feature mix only reached ≈ 0.09) and the elbow. Price sensitivity = conversion on ≥25% discounts minus conversion on ≤10% discounts.

Segment	Share	Days since last	Orders	Avg order value	Conversion	Discount lift
Occasional Buyers	29%	29	4.1	Rs 3,007	52%	+0.9%
Frequent Small-Basket Buyers	28%	29	11.2	Rs 2,694	54%	+13.7%
High-Value Regulars	26%	28	8.2	Rs 4,800	49%	+5.4%
Lapsing Customers	17%	70	6.9	Rs 3,609	49%	+2.3%

Frequent Small-Basket Buyers are the most price-sensitive segment - target them with coupons and bundles; win back **Lapsing Customers** with reminders; protect High-Value Regulars with service and loyalty perks.

5. GenAI: product RAG, policy QA and the controller

- **Product RAG:** one document per item (title, description, brand, category, attributes, price context, tags, review) → embeddings → FAISS (exact cosine search). Budget, sub-category, colour and brand are parsed by explicit rules and applied as hard filters (relaxed step by step, logged in the trace). Retrieval inspection on 300 dataset queries: sub-category precision@5 = 0.93, colour precision@5 = 1.00, exact item in top-5 = 0.66. The LLM must return JSON (recommended_products, explanation, supporting_ids); a validator drops any id not retrieved or over budget; empty / weak retrieval returns a clear no-match message.
- **Policy QA:** sentence-level policy chunks; hit@1 = 100% on 10 labelled questions. Answers must cite the policy_doc_id and quote it verbatim (validated); below the similarity threshold the assistant refuses.
- **Controller:** guardrails → rule-based intent scores with specificity rules → multi-intent split (≤3 tools, each gets its own clause) → LLM router only as fallback, validated against a fixed tool list. Tools: product_search, policy_qa, segment_insights, review_summary, summarise_products (session memory), calculator. Every step is printed in a trace.

Assistant test suite: 100% pass (LIVE (gpt-4o-mini + text-embedding-3-small))

id	query	routed_tools	checks_passed	PASS	latency_s
Q1	Recommend casual sneakers under Rs 4000 for daily wear.	product_search	3/3	True	0.43
Q2	Can I return a discounted product after 7 days?	policy_qa	2/2	True	0.003
Q3	Suggest products for a premium customer who prefers black formal shoes.	product_search	3/3	True	0.001
Q4	Which customer segment seems most price-sensitive?	segment_insights	1/1	True	0.0
Q5	Summarise the top three retrieved products and explain why they fit the query.	summarise_products	1/1	True	0.0
Q6	How long does express delivery take?	policy_qa	2/2	True	0.0
Q7	Is physical damage to my watch covered under the warranty?	policy_qa	2/2	True	0.0
Q8	Show me a blue backpack for travel under 3000 and tell me if I can exchange it later.	product_search,policy_qa	3/3	True	0.001
Q9	What do customers say about Nike running shoes?	review_summary	1/1	True	0.001
Q10	When will my refund reach me if I paid by UPI?	policy_qa	1/1	True	0.001
Q11	Recommend a red watch under Rs 300.	product_search	2/2	True	0.002
Q12	What is 15% off 3999?	calculator	1/1	True	0.0
Q13	What's the weather in Mumbai today?	(none)	1/1	True	0.0

Curated test queries with regression checks (notebook 06).

6. Evaluation

13/13 curated queries pass (routing accuracy 100%); checks include budget respected, correct sub-category/colour, ids grounded in retrieval, correct policy cited, quotes verbatim, out-of-scope refused. pytest regression suite: 23 passed in 1.64s.

Example queries and outputs

[Q1] Recommend casual sneakers under Rs 4000 for daily wear. → tools: product_search

```
Recommended products:
- [ITM0845] Skechers Black Sneakers Synthetic Shoes - Rs 2,745: Designed for daily wear and offers comfort and
style.
- [ITM0905] Adidas Black Sneakers Synthetic Shoes - Rs 2,414: Suitable for daily wear with a good average
rating of 4.3/5.
- [ITM0851] Nike Grey Sneakers Mesh Shoes - Rs 1,919: Offers comfort and style for daily wear with a solid
rating of 4.1/5.
Why: All recommended sneakers are under Rs 4000 and designed for daily wear, making them suitable for your
request. They come from reputable brands and have positive customer feedback regarding comfort and style.
```

[Q2] Can I return a discounted product after 7 days? → tools: policy_qa

```
Policy: It depends on the discount percentage. Returns are allowed within 7 days for unused products, but
discounted products can only be returned if the discount is 30 percent or less. [sources: POL_RETURNS]
```

[Q3] Suggest products for a premium customer who prefers black formal shoes. → tools: product_search

```
Recommended products:
- [ITM1341] Bata Black Formal Synthetic Shoes - Rs 3,850: This shoe is designed for office use and offers
comfort and style, making it suitable for a premium customer.
- [ITM0621] Puma Black Formal Leather Shoes - Rs 5,752: This premium leather shoe is designed for daily wear,
providing comfort and style, which aligns with the preferences of a premium customer.
- [ITM1288] Adidas Black Formal Leather Shoes - Rs 3,186: This shoe is designed for office use and has a
perfect rating of 5.0/5, indicating high customer satisfaction, appealing to premium customers.
Why: The recommended products include a mix of formal shoes from reputable brands like Bata, Puma, and Adidas,
all in black and suitable for office use. They cater to the premium segment with varying price points and high
customer ratings, ensuring quality and style for the shopper's need
```

[Q4] Which customer segment seems most price-sensitive? → tools: segment_insights

```
The most price-sensitive (largest conversion lift from deep discounts) segment is **Frequent Small-Basket
Buyers** (28% of interactions). Profile: avg age 38, avg order value Rs 2,694, avg item price Rs 2,415, avg
discount seen 18%, conversion 54% overall, 61% on >=25% discounts vs 48% on <=10% discounts (lift +14%).
```

[Q5] Summarise the top three retrieved products and explain why they fit the query. → tools: summarise_products

```
Summary of the top results:
- [ITM1341] Bata Black Formal Synthetic Shoes are a stylish option at a competitive price.
- [ITM0621] Puma Black Formal Leather Shoes offer a premium feel with a high rating.
- [ITM1288] Adidas Black Formal Leather Shoes are highly rated and provide excellent quality.
These products cater to a premium customer looking for stylish black formal shoes with varying price points
and quality.
```

7. Findings, limitations and next steps

- Conversion is driven by who the customer is (membership, income) and product quality (rating) more than by price; discounts pay off mainly for one behavioural segment - so discounting should be targeted, not blanket.
- Limitations: modest tabular signal (ROC-AUC ≈ 0.65-0.68); moderately separated segments; templated review text; unstable source ids; no product images.
- Next steps: richer behavioural data (sessions, prior views), a cross-encoder reranker, LLM-as-judge groundedness scoring, A/B tests of segment-specific offers.

Reproducibility: pip install -r requirements.txt; key in .env (OPENAI_API_KEY, optional OPENAI_BASE_URL for Vocareum); ./run_all.sh. Seeds fixed (42), temperature 0, LLM and embedding calls cached on disk; RETAIL_OFFLINE=1 runs everything without a key.